

PLASMA OBSERVATORY

Unveiling plasma energization and energy transport in the near Earth's plasma environment through multi-scale observations

Proposed by ESA as the M7 mission
Formal decision by the SPC at the next meeting in November 2026.

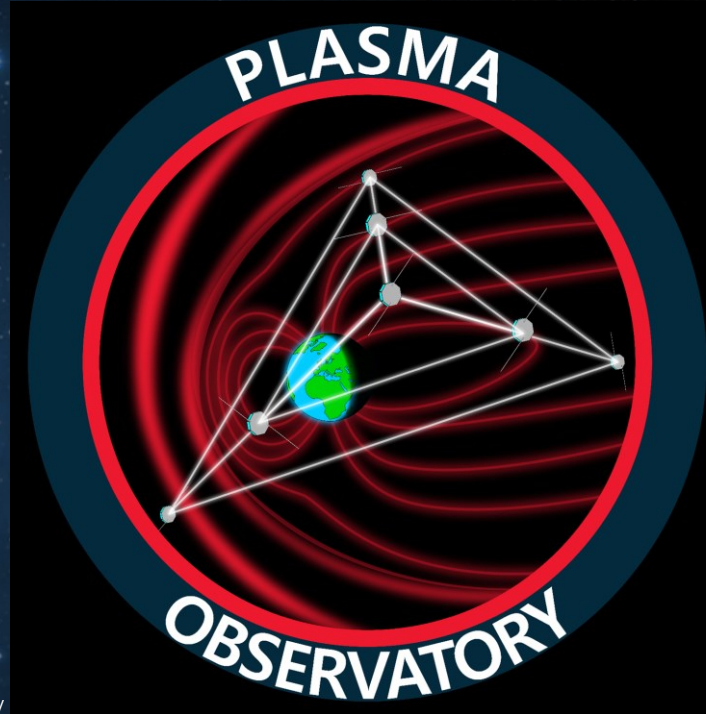
Targets two ESA Voyage 2050 themes for ESA-led M Mission:
Magnetospheric Systems and *Plasma Cross-scale Coupling*

Large scientific community: 450+ researchers from 25 countries (17 in Europe), including US, Japan and China

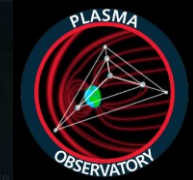
Payload team including 10+ ESA countries with key US and Japanese contributions

Next quantitative leap needed after Cluster and MMS to advance in both fundamental plasma and magnetospheric physics

Key component of the current international framework towards a new era of magnetospheric physics in the mid-2030s (e.g. NASA, JAXA)



Plasma Observatory DNA



SQ1: How are particles energized in space plasmas?

SQ2: Which processes dominate energy transport and coupling between the different regions of the Earth's Magnetospheric System?

Seven-point measurements to probe ion and fluid dynamics **simultaneously** in 3D in six key science regions of the Magnetospheric System.

HEO 7.2RE $\times$ 17RE, 15° inclination to cover key science regions under a variety of plasma conditions

Two tetrahedra sharing a vertex with variable inter-spacecraft separations, to cover all required ion and electron spatial scales in all key science regions

Electromagnetic field

B field: DC - 8 kHz

E field: DC to 100 kHz

Particles

Electrons and ion 3D VDFs @ 250 ms

H⁺, He⁺⁺, O⁺ 3D VDFs @ 2 s

Energetic electrons and ions VDFs @ 4 s

